

Email Classification System using NLP and Machine Learning

Overview

Organizations receive thousands of emails daily from customers, partners, and internal teams. Manually categorizing these emails is time-consuming, inefficient, and prone to human error. This project automates email classification using Natural Language Processing (NLP) and Machine Learning algorithms, enabling organizations to route emails to the appropriate departments quickly and accurately.

Problem Statement

Organizations receive a large volume of emails every day, including customer support requests, sales inquiries, complaints, feedback, and general communications.

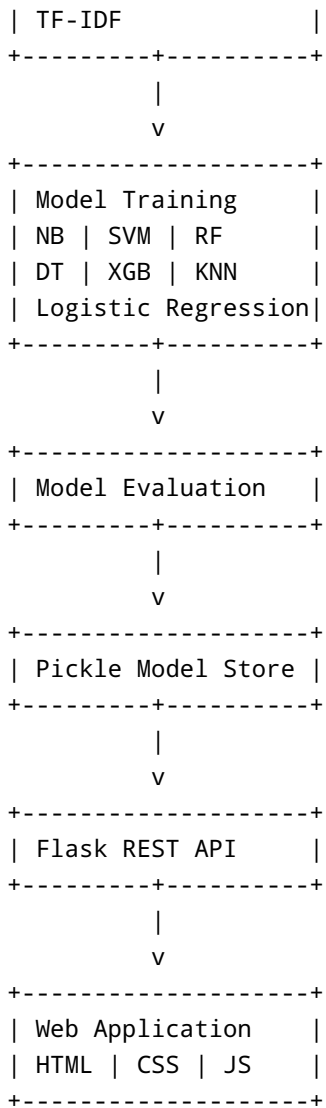
Manual classification of these emails:

- Consumes significant human effort
- Increases response time
- Leads to inconsistent categorization
- Reduces operational efficiency

The objective of this system is to automatically classify incoming emails into predefined categories using NLP and Machine Learning techniques, improving productivity and response management.

Architecture Diagram

```
+-----+
| Email Dataset |
+-----+
      |
      v
+-----+
| Text Preprocessing |
| Cleaning           |
| Tokenization       |
| Stopword Removal   |
+-----+
      |
      v
+-----+
| Feature Extraction |
| CountVectorizer    |
+-----+
```



End-to-End Workflow

1. Email Data Collection

- Collect labeled email datasets.
- Categories may include:
 - Support
 - Sales
 - Complaint
 - Inquiry
 - Feedback
 - Others

2. Text Preprocessing

- Lowercasing
- Removing punctuation

- Removing special characters
- Removing stopwords
- Stemming/Lemmatization

3. Tokenization

- Split email text into tokens (words).
- Prepare text for feature extraction.

4. Feature Engineering

CountVectorizer

Converts text into numerical vectors based on word frequency.

TF-IDF Vectorizer

Calculates term importance across documents and improves model performance.

5. Model Training

Multiple machine learning models are trained and compared:

- Naïve Bayes
- Support Vector Machine (SVM)
- Random Forest
- Decision Tree
- XGBoost
- K-Nearest Neighbors (KNN)
- Logistic Regression

6. Model Evaluation

Evaluate models using:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

The best-performing model is selected for deployment.

7. Model Serialization

- Save trained model using Pickle.
- Save vectorizer for inference.

8. REST API Development

Develop Flask APIs for:

- Email prediction
- Batch prediction
- Model health check

9. Frontend Development

Build an interactive UI using:

- HTML
- CSS
- JavaScript

Users can enter email content and instantly receive predicted categories.

10. Deployment

Deploy the solution as a Flask web application accessible through a browser.

Key Features

- Automated email categorization
 - NLP-based preprocessing
 - Multiple ML model comparison
 - Real-time classification
 - REST API integration
 - User-friendly web interface
 - Scalable architecture
 - Fast prediction response
-

Deployment Architecture

```
User
|
v
Frontend (HTML/CSS/JS)
|
v
Flask REST API
|
v
Pickle Model + TF-IDF Vectorizer
|
v
Prediction Engine
```

|
v
Email Category Output

Tech Stack

Programming Language

- Python

NLP

- NLTK
- CountVectorizer
- TF-IDF

Machine Learning

- Naive Bayes
- Support Vector Machine (SVM)
- Random Forest
- Decision Tree
- XGBoost
- K-Nearest Neighbors (KNN)
- Logistic Regression

Backend

- Flask
- REST APIs

Frontend

- HTML
- CSS
- JavaScript

Model Persistence

- Pickle

Data Processing

- Pandas
 - NumPy
-

Business Benefits

- Faster email routing

- Reduced manual effort
 - Improved customer response times
 - Increased operational efficiency
 - Better email management
 - Scalable enterprise solution
-

Future Enhancements

- Deep Learning-based email classification using LSTM and BERT
 - Multi-language email support
 - Email priority prediction
 - Auto-routing to departments
 - Integration with Outlook and Gmail
 - Explainable AI predictions
 - Cloud deployment on Azure, AWS, or GCP
-

Conclusion

The Email Classification System leverages Natural Language Processing and Machine Learning to automate email categorization, reducing manual effort and improving organizational productivity. The solution is production-ready with REST APIs, a web-based user interface, and support for multiple machine learning algorithms.